Finite-Sample Monitoring of BB84 Quantum Key Distribution under Channel Noise and Bounded Drift

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Abstract

I investigate the detection of restricted eavesdropping in BB84 quantum key distribution using synthetic public measurement counts. My study combines an explicit transmission-noise model, classical machine-learning baselines, finite-sample statistical tests and resource-matched calibration strategies. The central question is whether attack-induced changes can be distinguished from ordinary noise when the reference data and disclosed monitoring bits are finite. In corrected fixed-policy experiments, no learned classifier dominates across the tested attack families. I then analyze bounded honest drift and derive a model-specific overlap condition: a restricted intercept–resend attack with raw-bit knowledge floor κ can have the same complete public-count law as an allowed honest channel when $(1-\lambda)\kappa \le d_Z + d_X$. For $\lambda = 0.03$ and $\kappa = 0.01$, the equal-basis boundary is 0.485 QBER percentage points. In a retrospective comparison on 12,288 existing public-count pairs, a classical Berger–Boos construction improves detection over my conservative direct concentration rule; on the Plan-9 zero-drift probe, detection increases from 168/512 to 378/512. Sensitivity remains limited when nonzero honest drift is permitted. These results distinguish population non-identifiability from finite-sample conservativeness and show why observer assumptions, sampling costs and established statistical comparators are essential to evaluating QKD alarms. The analysis concerns public-count monitoring, not a complete secret-key security proof or a hardware demonstration.

1 Introduction

BB84 quantum key distribution (QKD) uses nonorthogonal quantum states to establish raw shared-bit candidates [1]. Statistical checks then inform further protocol decisions. In this project, I investigate what public measurement counts can reveal about restricted eavesdropping when ordinary noise, finite samples and reference-data costs are made explicit.

I initially compared learned classifiers, a quantum kernel and simple thresholds on synthetic attack labels. After correcting the physical noise model and strengthening the evaluation, I focused on a narrower question:

> With a finite trusted calibration block, a limited budget of publicly disclosed monitoring bits and a declared honest-noise family, which restricted attacks can a valid change alarm detect, and when are honest and attacked observations statistically indistinguishable?

This is a monitoring study, not a replacement for QKD security analysis. An approximately 11% QBER threshold from an idealized asymptotic treatment is not a universal intrusion alarm or a certificate for an arbitrary optical implementation. Error correction, authentication, parameter estimation and privacy amplification are not replaced by a classifier.

1.1 Objectives and scope

My objectives are to build a physically consistent simulation, compare alarms on common observations, quantify reference-data and disclosure costs, and identify when the declared honest and attacked observation families overlap. I use negative controls to separate model misspecification from genuine attack sensitivity. I also compare the proposed concentration-based rule with established classical inference rather than treating improvement over a weak baseline as evidence of a new statistical method.

1.2 Meaning of the principal quantities

QBER
Quantum bit error rate: disagreements among publicly compared, matched-basis bits divided by their comparison count.

Gain
Detection probability per emitted pulse at a specified source intensity.

False-alarm probability
Probability of alarming under an allowed honest law. Uniform control concerns every parameter admitted by that law family.

Power
Probability of alarming under a specified attack. Worst-case power is the infimum over a declared attack family, not a single selected probe.

Raw-bit knowledge proxy
Expected fraction of retained raw Alice bits known exactly by a restricted interception mechanism. It is not final-key leakage.

2 Physical and sampling model

2.1 States, gates and the transmission channel

Alice prepares $|0\rangle, |1\rangle$ in the Z basis or $|+\rangle, |-\rangle$ in the X basis. To prepare X-basis bit one, X precedes H. Bob changes basis with H when measuring in X and then measures in Z. With ideal preparation/measurement and no channel noise, matched outcomes are deterministic; conjugate-basis outcomes are equiprobable.

I place one explicit depolarizing channel between preparation and measurement:

$$\mathcal{E}_\lambda(\rho) = (1-\lambda)\rho + \lambda I/2, \qquad q_0 = \lambda/2. \tag{1}$$

Here ρ is the single-qubit density matrix, I the identity and λ a mixing parameter, not QBER. At $\lambda = 0.03$, honest matched-basis QBER is 0.015 with ideal local gates.

An earlier implementation attached channel-like noise after local gates, making state-dependent gate counts determine noise exposure. The corrected model separates transmission noise from optional local gate noise. Local device errors can create honest basis asymmetry and must not be attributed automatically to Eve.

Qiskit Aer density-matrix probabilities are checked against analytical/matrix references. The eight preparation/measurement cases use deterministic probability lookups up to numerical precision, rather than an unseeded finite-shot lookup. Local random generators and immutable bounded caches support reproducibility. Agreement with this model is not calibration against a real device.

2.2 Restricted intercept–resend

In the intercept–resend (IR) model, Eve intercepts with probability r, measures in Z with probability β, otherwise X, and resends her observed state before the downstream channel. With ideal local gates,

$$q_Z = q_0 + \frac{1-\lambda}{2} r(1-\beta), \tag{2}$$

$$q_X = q_0 + \frac{1-\lambda}{2} r\beta, \tag{3}$$

$$\overline{q} = q_0 + \frac{1-\lambda}{4} r. \tag{4}$$

Uniform bases make mean QBER independent of Eve's bias, although the per-basis disturbance changes. Full noiseless interception gives mean QBER 25%. This is an iid attack family, not a general coherent or collective adversary.

2.3 Photon counts, loss and restricted PNS

For weak coherent pulses with intensity ν, photon count k is Poisson. Effective per-photon transmission/detection is η:

$$P(k \mid \nu) = e^{-\nu}\nu^k/k!, \qquad G_\nu = 1 - e^{-\nu\eta}. \tag{5}$$

The illustrative settings are $\nu_s = 0.5$, $\nu_d = 0.1$, $\eta = 0.15$. The gain-matched photon-number-splitting (PNS) abstraction blocks single/vacuum pulses and forwards multiphoton pulses with one intensity-independent probability

$$u = \frac{1 - e^{-\nu_s\eta}}{1 - e^{-\nu_s}(1+\nu_s)}. \tag{6}$$

Infeasible $u > 1$ is rejected. Ideal forwarding preserves QBER but can change the decoy/signal gain ratio. This is not a complete multiphoton optical attack model or a full decoy-state security estimator. The latest drift research alternative is IR-focused, not a worst-case PNS guarantee.

2.4 Dataset and observation boundary

All evaluation data are synthetic. Pulse/session simulations and analytical aggregate-count samplers coexist; the large research campaigns do not execute a hardware circuit for each simulated emission.

My public-count detector receives six original public coordinates: error numerators and denominators for Z0, Z1, X0 and X1, plus signal/decoy detected and sent counts. Only randomly disclosed errors are public. Retained private bits, true noise probabilities, attack labels and Eve's knowledge do not enter the alarm. Public denominators are random and kept as observed, including zero-count rows. Selection is independent of errors under the declared model.

3 Methodology and implementation

I organize the implementation into simulation, observation construction, detector calibration and evaluation. The public observation boundary is fixed before computing an alarm: hidden simulator labels are used to score a decision, not to make it. Separate fitting and evaluation cohorts are used where specified by each experimental protocol.

Table 1: Implementation modules and responsibilities. Paths are relative to the `quantum-ai-lab` directory.

Component	Responsibilities
`src/bb84/`	Qubit/session model, synthetic datasets, public counts, honest nuisance and loss-attack abstractions.
`src/classical_ml/`	Logistic regression, random forest and SVM baselines; count/loss-aware detectors, frozen benchmarks and replay auditors.
`src/quantum_ml/`	Exact statevector fidelity kernel. No demonstrated quantum computational advantage.
`src/sequence_ml/` and `src/adversarial/`	Historical temporal CNN, sequence generation and constrained policy search. Corrected headline benchmark revalidation is incomplete for these legacy branches.
`src/research/`	Public calibration, identifiability, allocation and attack-search studies, stable-model power bounds, drift tests and classical audit.
`scripts/`, `tests/`, `docs/`	Repeatable runners, implementation checks, mathematical assumptions and saved protocol/results explanations.

The main numerical tools are NumPy, SciPy, Qiskit/Aer and scikit-learn. The final recorded statistical campaign used Python 3.12.14, NumPy 2.5.3 and SciPy 1.18.1. The simulation generators are not intended for cryptographic key generation.

Corrected evaluations separate fitting/calibration, attack-policy selection and evaluation where the individual protocol specifies fresh data. Comparisons use shared observations, measured honest alarm rates and explicitly frozen rules. Five-seed summaries use sample standard deviation, not confidence intervals. The classical audit is retrospective rather than a fresh holdout. A high area under the receiver operating characteristic curve (AUC) does not establish a useful low-false-alarm operating point.

4 Experimental design and baseline results

The following stages are distinct campaigns and cannot be combined as one dataset. I keep their outcomes separate because their observation windows, controls and detector calibration procedures differ.

Table 2: Distinct baseline and robustness campaigns. Counts refer to each campaign separately.

Stage	Recorded lesson
Fixed-policy comparison	29,500 sessions. Relative sensitivity depends on attack type; no universal ML advantage.
Frozen robustness	89,500 sessions. Honest noise/window shifts inflate false alarms for some frozen rules.
Count-aware monitoring	104,500 sessions. Exact tests improve selected honest controls but lose weak-attack sensitivity.
Honest loss/device stress	44,500 sessions. Changed honest loss exposes misspecified nominal-loss assumptions.
Loss-aware screening	137,000 rows. Signal-gain uncertainty improves tested loss robustness at a power cost.
Fixed-window sensitivity	70,000 sessions. More emissions improve selected power but amplify false alarms under a wrong device null.

I then examined honest/IR overlap, finite calibration, public sampling, resource allocation and constrained attack selection. The stable-noise conditional test and its continuous-family power analysis provide a reference for the later bounded-drift experiments. I use the classical-baseline comparison to test whether the direct rule's weakness reflects fundamental observational ambiguity or avoidable statistical conservativeness.

The corrected fixed-policy comparison gives the following mean rates across five seeds. Each entry is a percentage, with sample standard deviation after the ± sign; these are not confidence intervals.

Method	Honest alarms	Weak IR detection	Medium PNS detection
Calibrated QBER	5.5 ± 0.8	59.8 ± 2.6	5.4 ± 1.2
Decoy gain ratio	5.6 ± 0.5	5.0 ± 1.1	29.0 ± 3.7
Joint max-tail	5.0 ± 1.0	49.7 ± 1.8	14.7 ± 2.0
Random forest	5.7 ± 0.7	42.2 ± 3.7	25.1 ± 3.2

Table 3: Corrected fixed-policy baseline results. Rates belong to the 29,500-session campaign, with 6,000 emissions per session. Attained honest alarm rates differ slightly, so this is not an exact matched-size comparison.

Table 3 shows complementary sensitivities rather than a universal ML advantage. QBER is stronger for the selected weak IR attack; gain-ratio information is more useful for the selected PNS attack. In the separate frozen robustness study, random-forest honest alarms increase from 4.6% to 20.1% under higher honest noise. In a device-noise negative control, the joint rule reaches 67.84% honest alarms under a misspecified common-basis null. More observations cannot repair the wrong null model. Quantum-kernel and temporal experiments are exploratory and are not used as validated superiority claims in this report.

5 Finite reference data, public disclosure and attack constraints

Let B be total emissions, M calibration emissions and f the independent monitoring disclosure probability. All reference test observations are sacrificed. Define emission-equivalent sacrifice exposure

$$C = M + f(B - M), \qquad R = (1 - f)(B - M) = B - C. \tag{7}$$

Under common gain/sifting laws, equal C matches expected sacrificed counts; R similarly matches retained-candidate exposure. Neither is itself a secret-key length, and realized counts remain random. Matching expectations does not match latency, energy or distilled secure-key yield.

The main plans have $B = 4{,}000{,}000$ and $C = 2{,}000{,}000$:

Plan	Calibration emissions	Monitoring emissions	Disclosure f
9	800,000	3,200,000	0.375
10	1,333,333	2,666,667	666,667/2,666,667

Table 4: Calibration/disclosure allocations with equal expected sacrifice exposure. Plan identifiers follow the saved experimental protocol.

At the illustrative gain setting, each has approximately 43,572 expected sacrificed observations and the same expected retained candidates. Signal and decoy sifted observations are included in this candidate-pool accounting.

Eve's restricted constraint is

$$I_{\text{cert}} = r/2 \geq \kappa, \qquad r \in [2\kappa, 1], \quad \beta \in [0, 1]. \tag{8}$$

This counts expected exactly-known retained raw Alice bits from basis-matched interceptions under uniform independent selection. It excludes chance-correct guesses and does not equal mutual information, min-entropy or final-key leakage. The 1% floor means $\kappa = .01$ and minimum interception $r = .02$.

5.1 Stable-model continuous power brackets

The earlier stationary conditional exact detector was analyzed by conditioning on random public denominators/gains and integrating binomial errors. A contained increased-error rejection event is monotone even though full two-sided power is not assumed monotone. Bias cells and reflection cover the restricted continuous family, not merely sampled policies.

The following bracket the continuous minimum power at two fixed plans on the declared at-least-95% joint Monte Carlo coverage event, subject to floating-point limits:

Raw-knowledge floor	Plan 9 minimum power	Plan 10 minimum power
0.25%	2.790–10.276%	2.747–10.157%
0.5%	12.202–19.837%	11.975–19.516%
1%	54.376–62.809%	52.774–61.139%
2%	97.721–100%	97.565–100%

Table 5: Continuous-family minimum-power brackets for the stationary conditional detector. These do not apply to the bounded-drift detectors.

The upper endpoint comes from an admissible balanced candidate, not an exact optimizer. Small positive lower bounds need not provide useful excess power over honest alarms. These are **stable-noise** results for a different detector; they do not certify either of the latest drift rules. Exact balanced optimality and a global allocation optimum remain unresolved.

6 Bounded honest drift and population indistinguishability

For basis $b \in \{Z, X\}$, the independent calibration and monitoring blocks have constant error probabilities θ_b and μ_b, shared by both preparation bits in each basis:

$$0 \leq \theta_b, \mu_b \leq 1/2, \qquad |\mu_b - \theta_b| \leq d_b. \tag{9}$$

The allowances d_b are absolute probability differences declared before monitoring. They are sensitivity assumptions, not measured device drift. This rectangle has admissible honest Pauli representatives, but that fact does not establish that a particular observed channel is innocent.

Gains and intensities are stable; selection is error-independent. Calibration is assumed genuinely attack-free. Authentication of classical messages does not establish that the quantum reference channel was free of Eve.

Importantly, the current *attack* branch retains fixed downstream noise $q_0 = \lambda/2$. It does not cover an attack on concurrently drifting noise. The question is whether that fixed-baseline attack can masquerade as allowed honest drift.

6.1 Overlap condition

Write $c = (1 - \lambda)/2$. Attack excess errors satisfy

$$\Delta_Z = cr(1 - \beta), \quad \Delta_X = cr\beta, \quad \Delta_Z + \Delta_X = cr. \tag{10}$$

An attack fits inside the honest rectangle exactly when its two nonnegative excesses fit the corresponding allowances. A feasible bias exists iff $cr \leq d_Z + d_X$. Minimizing allowed r therefore gives the existence criterion

$$\boxed{(1 - \lambda)\kappa \leq d_Z + d_X.} \tag{11}$$

For equal allowances, $d_* = (1 - \lambda)\kappa/2$. At $\lambda = .03$, $\kappa = .01$, this is .00485, or **0.485 QBER percentage points per basis**.

With positive error-observation resources in both stages, the matched honest and attacked laws agree in their calibration, gain, selection and conditional binomial-error distributions. The complete public-count laws agree, not only their expected QBER. If an alarm has honest false-alarm probability at most α, it has power at most α on the matching attack. Thus worst-case power at/above overlap is at most α using this same observation protocol, irrespective of more emissions. Absent calibration, zero transmission or no monitoring error disclosure introduce additional degeneracies.

This is a model-scoped application of statistical identifiability. It does not imply a failure of the BB84 security proof. Outside overlap, finite-sample power can still be poor, but poor power of one rule is not a universal impossibility result.

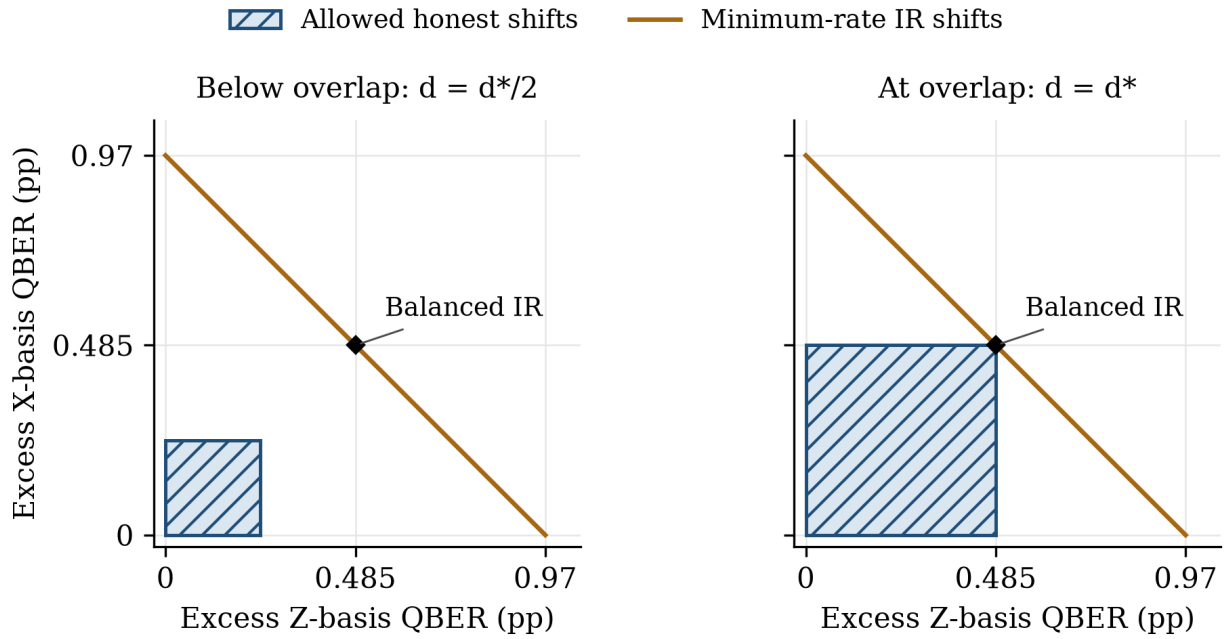


Figure 1: Geometry of honest/IR overlap for $\lambda = 0.03$ and $\kappa = 0.01$. The shaded square shows the positive-quadrant part of the honest drift region; the line shows minimum-rate IR excess errors with total 0.97 QBER percentage points. At $d = d_* = 0.485$ percentage points, balanced IR enters the honest region. This figure illustrates the model geometry, not empirical drift data.

7 Finite-sample detectors and classical comparison

7.1 Confidence-gap and direct concentration rules

The baseline forms simultaneous Hoeffding or Clopper–Pearson (CP) intervals for the probabilities at each stage and coordinate. It alarms when their gap exceeds the allowed change. Empty counts give [0,1]. The direct rule pools Z0/Z1 and X0/X1 under their shared probabilities, retaining separate signal/decoy gains.

For one pooled basis, let Y_c, Y_m be public error counts and n_c, n_m their denominators. With $D = Y_m/n_m - Y_c/n_c$, the direct rule uses a one-sided CP upper bound on the unknown variance and a Bernstein radius

$$h = bL/3 + \sqrt{2\widehat{V}L + (bL/3)^2}, \qquad b = \max(1/n_c, 1/n_m), \quad L = \log(16/\alpha). \tag{12}$$

It checks both error directions and both gain directions with an explicit union budget. Validity is conditional on the model and in real arithmetic. Pooling is not justified under arbitrary bit-dependent noise or selection.

The direct-rule experiment used 110,592 fresh public pairs to compare this rule with older confidence-gap baselines. Its Plan-9 zero-drift balanced 1% probe gave direct 168/512 alarms versus unpooled CP 22/512. That improvement against a conservative baseline did not establish a new method or adequate weak-attack protection.

7.2 Established classical comparator

I recognize the per-basis core as nonzero-margin two-binomial inference, a longstanding statistical problem [2, 3]. I therefore add a Berger–Boos confidence-set construction [4], not a new inequality or a claim to reproduce Chan's full score-ordered test.

For the upward direction, the null is $p_m - p_c \le d$. Let $a = \alpha/8$, $\gamma = a/10$, and C be a two-sided CP confidence set for p_c, intersected with [0,.5]. Define

$$p_{\mathrm{BB}} = \min\left\{1, \gamma + \sup_{p\in C} \Pr_{p,\min(.5,p+d)}(T \ge T_{\mathrm{obs}})\right\}, \quad T = Y_m/n_m - Y_c/n_c. \tag{13}$$

On confidence coverage, the supremum includes the true calibration parameter's boundary-tail p-value. Coverage failure costs γ; the remaining rejection event costs at most $a - \gamma$. No independence of C and T is required.

For a parameter cell [l,u], the tail at calibration probability l and monitoring probability $\min(.5, u + d)$ dominates the tail throughout that cell by monotone coupling. This supplies continuum upper envelopes rather than treating a parameter grid as exact maximization. At most 64 subdivisions are allowed; unresolved directions do not reject. Integer-inclusive thresholds and explicit omitted-mass bounds protect the finite-support calculation in real arithmetic. Ordinary SciPy evaluations are not outward-rounded certificates.

The downward direction exchanges stages. Four error directions spend $\alpha/2$ and the unchanged gain directions the other $\alpha/2$. Both finite-sample rules therefore have the same model-conditional upper-α ceiling, not equal attained false-alarm rates. An additional restricted-score normal-tail diagnostic is explicitly approximate and is not assigned that finite-sample guarantee.

7.3 Retrospective replay findings

The bounded audit replays all 512 saved pairs in 24 selected Step-12 cohorts: two plans, the 1% raw-bit floor, zero/half-overlap allowances, five honest controls and one balanced floor-rate attack.

There are 12,288 existing pairs; all detector decisions are evaluated on common saved observations.

Plan	Allowance	Direct	Classical BB	Approx. score
9	Zero	168/512 (32.81%)	378/512 (73.83%)	415/512 (81.05%)
9	Half boundary	12/512 (2.34%)	79/512 (15.43%)	107/512 (20.90%)
10	Zero	151/512 (29.49%)	353/512 (68.95%)	394/512 (76.95%)
10	Half boundary	18/512 (3.52%)	76/512 (14.84%)	120/512 (23.44%)

Table 6: Alarm counts and detection percentages for the balanced floor-rate IR probe. BB denotes Berger–Boos. The score diagnostic is approximate and does not share the finite-sample guarantee of the two other rules.

Half the boundary here is 0.2425 QBER percentage points per basis. The four paired BB-minus-direct gains, in percentage points, are:

Plan	Allowance	Gain	Nominal paired joint interval
9	Zero	41.02	[31.97,48.76]
9	Half boundary	13.09	[6.85,18.95]
10	Zero	39.45	[30.48,47.17]
10	Half boundary	11.33	[5.43,16.92]

Table 7: Paired BB-minus-direct detection differences in percentage points. Intervals use nominal joint fixed-analysis error allocation across saved rate/effect summaries; they do not adjust for retrospective research selection.

The largest observed honest counts across 20 selected controls are 2/512 for direct, 11/512 for BB and 16/512 for approximate score. These measured rates differ; low measured alarms alone do not prove uniform validity.

The 72 rate intervals and 24 paired effects have explicit nominal fixed-analysis failure-budget accounting. Because scope and comparator design are retrospective, they are not independent confirmation of an unrestricted adaptive research process. Fixed-probe gains do not establish uniform dominance or continuous worst-case power. Even 15% detection at half the boundary misses most trials.

Interpretation. Standard statistical machinery recovers a large sensitivity gap. The direct rule's weakness outside population overlap was partly conservativeness, not an established universal detection limit. I interpret this improvement as evidence for a stronger established baseline, not a new statistical inequality.

8 Discussion

8.1 Interpretation and relation to previous work

My results separate three effects. First, increasing honest noise can invalidate a threshold calibrated for a narrower channel model. Second, a conservative finite-sample rule can lose substantial power even where the observation families remain distinguishable. Third, when honest and attacked public-count laws overlap, changing the classifier cannot restore information absent from the observations. Figure 1 addresses the third effect; Figure 2 addresses the second. They are not interchangeable claims about a fundamental detection limit.

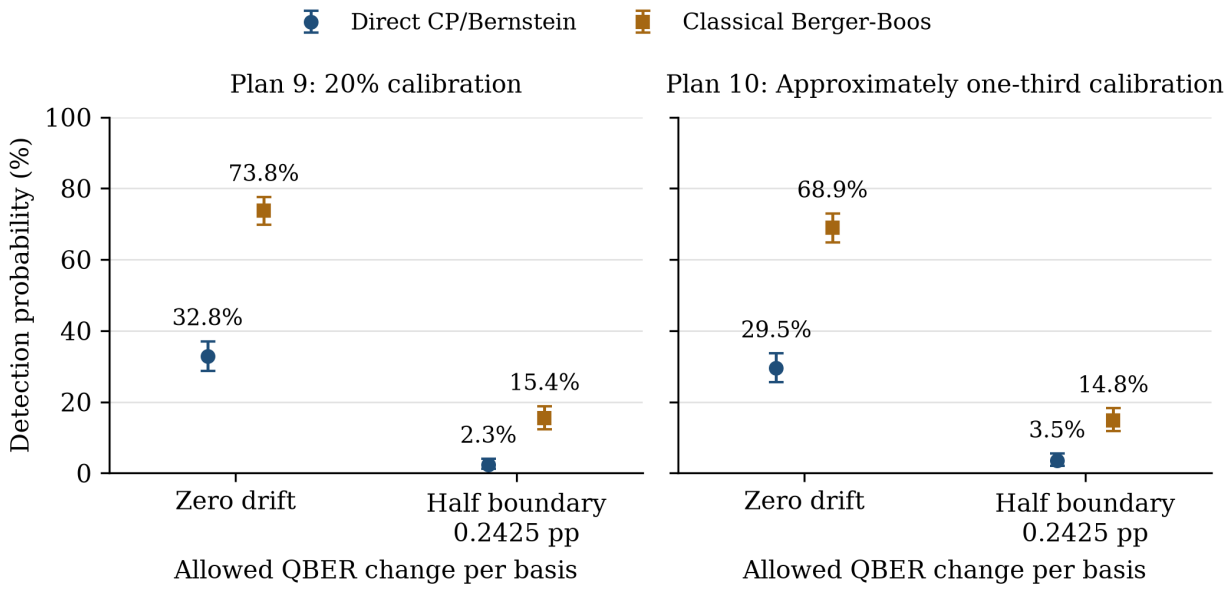


Figure 2: Detection of the balanced floor-rate IR probe in the existing-data comparison. Each point uses 512 paired trials; error bars are nominal marginal 95% binomial intervals, not selection-adjusted simultaneous bands. Both rules have a model-conditional 5% upper false-alarm ceiling, but their attained honest alarm rates differ.

BB84 eavesdropping detection and noisy partial intercept–resend are established research topics [5, 6]. Nonzero-margin binomial comparisons and confidence-set nuisance maximization are established statistical methods [2, 3, 4]. Finite-statistical tools also play a broader role in QKD parameter estimation [7]. I therefore frame this work as an implementation and evaluation of restricted public-count monitoring, including its identifiability and resource limitations, rather than proposing a new quantum effect or replacing existing security analyses.

8.2 Scientific limitations

1. No empirical device dataset or hardware validation establishes the scale or temporal structure of the drift allowances. Dark counts, afterpulsing, dead time and realistic source/detector fluctuations are not covered generally.

2. Attack-free calibration is a substantive trust assumption. The current attacked branch does not include concurrently drifting background noise, calibration poisoning, arbitrary adaptive strategies or coherent attacks.

3. Public-count monitoring is not error attribution in an unrestricted model. Standard QKD security decisions need not identify the cause of every error. No benefit in composable secret-key yield, operational cost or availability has been established for this added monitoring layer.

4. The raw-bit proxy is not a final-key compromise measure. Full reconciliation, privacy amplification, entropy estimation and composable security accounting have not been implemented by these studies.

5. Continuous worst-case power for the latest drift rules, optimal allocations and balanced Eve's

exact optimality remain unresolved. The bounded classical audit is not an exhaustive comparison of all exact margin tests.

6. Numerical validity is distinguished from a real-arithmetic argument. The implementation uses ordinary floating-point distribution functions, not outward-rounded interval arithmetic. Retrospective comparisons do not provide an independent confirmation of adaptively selected research findings.

8.3 Future work

Useful extensions would connect the declared drift allowances to device measurements, model attacks on concurrently drifting channels and evaluate operational consequences such as unnecessary aborts or retained secure-key yield. Additional detector comparisons should match attained false-alarm rates where possible and separate algorithm design from independent evaluation. Sequential monitoring would also require explicit control of repeated testing; reusing fixed-window thresholds indefinitely would not supply that control.

9 Conclusion

I developed and evaluated a simulation-based framework for studying restricted BB84 eavesdropping alarms under channel noise, finite calibration and public sampling constraints. The corrected baseline experiments show attack-specific trade-offs rather than a universal advantage for learned classifiers. The bounded-drift analysis identifies a precise condition under which an eligible IR attack and an allowed honest channel generate identical public-count laws. Below that overlap boundary, the retrospective classical comparison shows that established nuisance-parameter inference can recover much of the power lost by a conservative concentration rule. My main conclusion is that an alarm must be evaluated together with its observation protocol, honest-noise family and resource budget. Good classification on synthetic labels alone does not establish reliable intrusion attribution or secure final-key generation.

A Implementation verification and reproducibility

The documented full regression run passed 896 tests in 64.33 seconds, including 27 tests for the classical comparison. The replay checker reconciles 12,288 selected original pairs, 72 unique rate cells and 24 unique paired effects. Source and archive fingerprints are retained to distinguish a detector replay from regeneration of the underlying observations.

Independent calculation paths include finite-table sums, reversed full-support binomial integration, CP interval inversion and near-threshold saved-row checks. Some checks share numerical libraries, so agreement is not an independent real-device validation. Finite grids do not replace the uniform mathematical argument.

I retain all 2,048 zero-transmission rows in the comparison. Of 49,152 error direction evaluations, three exhaust the nuisance-envelope budget and return nonreject. No gain alarms occur in this selected subset; the gain branch is retained and checked separately rather than empirically exercised by these particular cohorts.

The accompanying repository retains the physical assumptions, drift model and comparison protocol under `quantum-ai-lab/docs/`. Numerical summaries, paired decisions and verification records are stored in `quantum-ai-lab/results/benchmarks/classical-drift-audit-v1/`.

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